

Day 12 - Advanced Task: Student Course Enrollment System

Goal:

Build a system where a course has many students and each student belongs to one course.

Requirements:

1. Database Design

Courses table:

- id
- name
- duration

Students table:

- id
- name
- email
- course_id (foreign key)

2. Relationships

- Student belongsTo Course
- Course hasMany Students

3. Features

- Create Course (add + list)
- Create Student with course dropdown (dynamic from DB)
- Validate name (required), email (required + unique)
- Store student with course_id

4. Student Listing

Display:

Student Name | Email | Course Name | Duration

5. Data Fetching

Use Eloquent:

```
Student::with('course')->get();
```

6. Filtering

Filter students by course

7. Bonus

- Show total students per course
- Course details page with students list
- Handle empty courses (no students)

Questions:

- Why belongsTo in Student?
- What is N+1 problem?
- Why use with()?